Logic Homework 2

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Question 1

Let: $\varphi = \forall x.P(x) \vee R(x)$, $\psi = ((\forall y.P(y)) \vee (\exists z.R(z)))$. We show that: $\varphi \vDash \psi$:

$$
\begin{aligned}
[[\varphi]] = 1 &\iff [[R(x)]]v[x \longmapsto a] = 1 \text{ or } [[P(x)]]v[x \longmapsto a] = 1 \text{ for all } a \in |\mathcal{M}| \\
&\iff [[P(x)]]v[x \longmapsto a] = 0 \text{ then } [[R(x)]]v[x \longmapsto a] = 1 \text{ for all } a \in |\mathcal{M}| \quad \text{(1a)} \\
&\iff [[R(x)]]v[x \longmapsto a] = 0 \text{ then } [[P(x)]]v[x \longmapsto a] = 1 \text{ for all } a \in |\mathcal{M}| \quad \text{(1b)}
\end{aligned}
$$

The semantic entailment is resolved by the following:

$$
\begin{aligned}
&[[\varphi]]v(x) = [[\psi]]v(y,z) = 1 \\
&\iff [[\forall y.P(y)]]v(y) = 1 \text{ or } [[\exists z.R(z)]]v(z) = 1
\end{aligned}
$$

Then we go over all the cases:

1) if $[[\forall y.P(y)]]v(y,z) = 1$, then:

$$
\begin{aligned}
[[\forall y.P(y)]]v(y,z) &= \min\{[[P(y)]]v[y \longmapsto a] | a \in |\mathcal{M}|\} = 1 \\
&\iff [[P(y)]]v[y \longmapsto a] = 1 \text{ for all } a \in |\mathcal{M}| \text{ making } [[\psi]]v[y \longmapsto a] = 1
\end{aligned}
$$

2) if $[[\forall y.P(y)]]v(y) = 0$, then:

$$
\begin{aligned}
[[\forall y.P(y)]]v(y) &= \min\{[[P(y)]]v[y \longmapsto a] | a \in |\mathcal{M}|\} = 0 \\
&\iff [[P(y)]]v[y \longmapsto a] = 0 \text{ for some } a \in |\mathcal{M}| \text{ then } [[R(y)]]v[y \longmapsto a] = 1 \\
&: \text{ by assumption 1a, making } [[\psi]]v[y \longmapsto a] = 1
\end{aligned}
$$

3) if $[[\exists z.R(z)]]v(z) = 1$, then:

$$\begin{aligned}[[\exists z.R(z)]]v(z) &= \max\{[[R(z)]]v[z \longmapsto a]|a \in |\mathcal{M}|\} = 1 \\ &\iff [[R(z)]]v[z \longmapsto a] = 1 \text{ for some } a \in |\mathcal{M}| \text{ making } [[\psi]]v[z \longmapsto a] = 1\end{aligned}$$

4) if $[[\exists z.R(z)]]v(z) = 0$, then:

$$\begin{aligned}[[\exists z.R(z)]]v(z) &= \max\{[[R(z)]]v[z \longmapsto a]|a \in |\mathcal{M}|\} = 0 \\ &\iff [[R(z)]]v[z \longmapsto a] = 0 \text{ for all } a \in |\mathcal{M}| \text{ then } [[P(y)]]v[y \longmapsto a] = 1 \\ &: \text{by assumption 1b, making } [[\psi]]v[z \longmapsto a] = 1\end{aligned}$$

We've exhausted all cases and all cases lead to $[[\varphi]]v(x) = [[\psi]]v(y, z) = 1$, therefore $\varphi \vDash \psi$.

Question 2

Let: $\varphi = \forall x.\forall y.P(x, y) \to P(y, x)$, $|\mathcal{M}| = \mathbb{N}$, $\mathcal{R} = \{P(m, n) = m \doteq n\}$, $\text{ar}(P) =$ 2.

We show that φ is satisfiable:

$$\begin{aligned}[[\varphi]]v(x) &= \min\{[[\varphi]]v[x \longmapsto a, y \longmapsto b]|a, b \in |\mathcal{M}|\} = 1 \\ &= \min\{[[P(x, y)]]v[x \longmapsto a, y \longmapsto b] \implies [[P(y, x)]]v[x \longmapsto a, y \longmapsto b]|a, b \in |\mathcal{M}|\} = 1 \\ &= \min\{\mathcal{M}(P)(a, b) \implies \mathcal{M}(P)(b, a)|a, b \in |\mathcal{M}|\} \\ &= \min\{a \doteq b \implies b \doteq a|a, b \in |\mathcal{M}|\} \\ &= 1 : \text{shown by the symmetry property of equality.}\end{aligned}$$

This shows that $\vDash_M \varphi$, therefor φ is satisfiable.